SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B. TECH DEGREE 2nd SEMESTER SECOND INTERNAL EXAMINATIONS, JULY 2024

EC B/IT B 23-200 - 0201B : LINEAR ALGEBRA & TRANSFORM TECHNIQUES (2023 Scheme)

TIME: 2 hrs **MAX. MARKS: 30**

Course Outcomes

On completion of this course the student will be able to:

CO1: Solve linear system of equations and to determine eigen values and vectors of a matrix.
CO2: Exemplify the concept of vectorspace and it's subspace.
CO3: Determine Fourier series expansion of functions and transform.
CO4:Solve linear differential equation and integral equation using Laplace transform.

PART A

(Answer all questions) **(5 x 2= 10 Marks)**

		Marks	BL	CO	PO
I (a)	Define vector space with an example.	2	L1	2	1
(b)	Write the standard basis and dimension of m*n matrix over a field F.	2	L2	2	2
(c)	Show that the set of all 2*2 non singular matrices is not a subspace.	2	L3	2	2
(d)	Prove that the set {(1,2), (1,3)} form a basis for R^2.	2	L1	3	3
(e)	Find the Fourier series representing x in the interval $[-\pi,\pi]$	2	L3	3	2

PART B

(Answer the following questions)
(2 X 10= 20 Marks)

		Marks	BL	CO	PO
II (a)	Let V be a vector space of all 2*2 matrices . Prove that dim V =4	5	L2	2	3
(b)	Determine whether the vectors (1,3,2) ,(1,-7,8) , (2,1,-1) are linearly independent or dependent.	5	L2	2	2
	OR				
III (a)	Let R^3 be the inner product space with standard inner product then find normal of (3,0,4) and normalize (1/2, -1/3, 1/4).	5	L3	2	3

	(b)	If u=(cos x, sinx) v=(-sinx, cosx) then prove that {u,v} form an orthonormal basis for R^2.	5	L3	2	2
IV	(a)	Find the Fourier cosine series which represents f(x)= π-x in $0<x<\pi$	5	L3	3	3
	(b)	Assuming convergence for the above series at x=0 , deduce that $\pi^2/8 = 1+1/3^2 + 1/5^2$......	5	L3	3	3
		OR				
V	(a)	If f(x)= 1. 0<x<a 0. x>a Find Fourier sine transform.	5	L2	3	2
	(b)	Find Fourier cosine transform.	5	L2	3	2

Bloom's Taxonomy Levels

L1 - 16% L2- 43% L3-41%